

# Gideon A. Ong

ga09819@utulsa.edu | +1 (918) 409-2444 | gideonong.me | linkedin.com/in/gideon-a-ong

## Education

### The University of Tulsa, Tulsa, OK

*In progress*

International Engineering and Language Program (IEL)

Mechanical Engineering, B.S.M.E.; Spanish, B.A.; International Business, B.S.B.A.

GPA: 4.0

## Skills

Programming: Python, MATLAB, Java, R, JavaScript/Google Apps Script, Mathematica, Stata

Software: SolidWorks, PolyWorks, QGIS, Excel, Google Sheets, GitHub Copilot, GIMP, Inkscape

Certifications: CSWP, GitHub Copilot, Certiport Java Specialist, McGraw-Hill Excel Black Belt – Level 4

Machine shop: CNC and manual mills and lathes, MIG welding, saw, sander, 3D printing

Languages: English (fluent), Spanish (fluent), French (beginner)

## Experience

### Engineering Intern, ALS R&D · Baker Hughes, Claremore, OK

*Summer 2026*

- Co-built Velora with Zohaib Sheikh: a Python app that turns dyno test data into finished reports, cutting 8–40 hour report times by 80%+ (worth an estimated \$200k a year).
- Built AutoScan, a Python app that automates a KEYENCE 3D optical scanner so scan batches run unattended, saving about 10 hours of operator time per batch (at least \$5k a year in labor).
- Presented my GitHub Copilot workflow to the R&D team, gave peer-to-peer training, and wrote lessons-learned documents; was asked to write a white paper on the workflow.

### Machine Shop Technician · McElroy Prototyping Lab, The University of Tulsa

*Spring 2026 – present*

- Machine parts on CNC and manual mills and lathes, mostly aluminum for student projects, plus some plastics and other materials for student clubs and research groups.

### Undergraduate Research Assistant, TURC · Dr. Janica Edmonds, The University of Tulsa

*Summer 2025*

- Designed a Math Teachers' Circle session on map projections, a topic new to the circle, and led it with Tulsa-area teachers in October 2025.

## Projects

### Paint Sprayer Pump Cylinder Failure Analysis · ME 3033 Properties of Materials

*Spring 2026*

- Traced cracked pump cylinders on a client's paint sprayers (\$10–15k per failure) to rinse water freezing inside, using hardness testing, microstructure analysis, and thick-wall stress analysis.
- Served as technical lead for the prototype (overall team lead: Rachel Lewis); co-designed and machined a spring-loaded pin that lifts the valve so the cylinder drains in about 20 seconds, as a \$135 retrofit.
- Won the 2026 Hackworth-Willson Prize for Excellence in Failure Analysis.

### Chilled Water Pipeline Design · ES 3003 Fluid Mechanics

*Fall 2025*

- Routed and sized a replacement chilled-water pipeline for TU's campus in Python, using AI-assisted convex-hull routing and a separate pipe-diameter optimization; it scored 100% with the lowest-cost bid in the class.

### Stress Relief for a Notched Beam · ES 3023 Mechanics of Materials

*Fall 2025*

- Used SolidWorks simulation to design a spline cutout that reduced a notched beam's stress concentration factor by 42% (1.39 vs. 2.38) in four-point bending tests.

### Predicting Bicycle Crash Severity · ES 4863 AI for Engineers

*Spring 2025*

- Tested FAMD and undersampling to predict UK bicycle crash severity (about 40,000 crashes, 70 features), using SHAP to trace results back to the original variables.

### Recurring Crash Hotspots in the UK · ES 4863 AI for Engineers

*Spring 2025*

- Found UK intersections that stayed crash hotspots year after year by running DBSCAN on 1.6 million accident records (2005–2014), mapped them in QGIS, and confirmed them against local news reports.

### Eagle Scout Project: Wheelchair-Accessible Pathway · The Bee Community, Bryan, TX

*Summer 2022*

- Planned and led 27 volunteers (about 285 total hours) to build a 3-ft-wide, 273.5 sq ft ADA-compliant decomposed granite pathway for a business that employs adults with special needs.